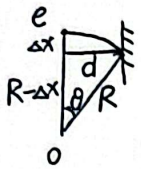
18.4



电子离开电子枪的速度 v

$\frac{1}{2}mv^2 = eU$

解 $v = \sqrt{\frac{2eU}{m}} = \sqrt{\frac{2\times1.6\times10^{-19}\times2\times10^4}{9.1\times10^{-31}}} = 8.38\times10^7\ m/s$

电子在磁场作用下运动轨迹半径

$\frac{mv^2}{R} = evB$ 解 $B = \frac{mv}{eB} = \frac{1}{B}\sqrt{\frac{2mU}{e}} = \frac{1}{0.5\times10^{-4}}\sqrt{\frac{2\times9.1\times10^{-31}\times2\times10^4}{1.6\times10^{-19}}}$

$= 9.54\ m$

电子偏转距离 Δx

$(R-\Delta x)^2 + d^2 = R^2$

解 $\Delta x = R - \sqrt{R^2-d^2} = 9.54 - \sqrt{9.54^2-0.2^2} = 2.1\times10^{-3}\ m$

偏转角度 θ

$\theta = \arcsin\frac{d}{R} = \arcsin\frac{0.2}{9.54} \approx 1.2°$

故偏转不显著

18.12 (1) 因 $U_{AA'} = 6.55\ mV$

即 A侧电势高于A'侧电势

结合电流 I 与磁场 B 方向

易知此半导体是负电荷导电(N型)

(2) 载流子浓度 n

$e\frac{U_{AA'}}{b} = Bev$

$I = n\cdot a\cdot b\cdot v\cdot e$

解 $n = \frac{IB}{U_{AA'}\cdot e\cdot a} = \frac{1\times10^{-3}\times3000\times10^{-4}}{6.55\times10^{-3}\times1.6\times10^{-19}\times10^{-3}} = 2.86\times10^{20}\ 个/m^3$

18.18 (1)

$\boxed{SI}$

线圈磁矩 $m = NI\cdot S = 200\times8.0\times(150\times10^{-3})^2$

$= 36\ A\cdot m^2$

(2) 作用在线圈上的力矩

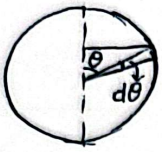
$\vec{M} = \vec{m}\times\vec{B}$

则 $M_{max} = mB = 36\times4 = 144\ N\cdot m$

扫描全能王 创建

18.21 电球壳单位面积上的电量σ

$e=4\pi R^2\sigma$



$$m=\int dI\cdot S=\int\frac{dq}{T}\cdot S$$
$$=\int_0^{\pi}\frac{\sigma\cdot ds'}{T}\cdot S$$
$$=\int_0^{\pi}\sigma\frac{R\cdot 2\pi R\sin\theta}{\frac{2\pi}{w}}\cdot\pi(R\sin\theta)^2d\theta$$
$$=\int_0^{\pi}\sigma\pi R^4w\sin^3\theta\,d\theta$$
$$=\int_0^{\pi}\frac{e}{4\pi R^2}\pi R^4w\sin^3\theta\,d\theta$$
$$=\frac{we}{4}R^2\int_0^{\pi}\sin^3\theta d\theta=\frac{eWR^2}{2}\int_0^{\frac{\pi}{2}}\sin^3\theta d\theta=\frac{eWR^2}{2}\times\frac{2}{3}=\frac{eWR^2}{3}$$

电子赤道上的线速度

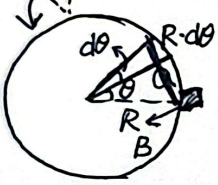
$$V=WR=\frac{3m}{eR}=\frac{3\times0.928\times10^{-23}}{1.6\times10^{-19}\times10^{-18}}=1.74\times10^{14}\ m/s>3\times10^8\ m/s$$

V已超越光速

故此经典模型不合理

18.23

取长为dl，宽为dt的壁筒进行分析

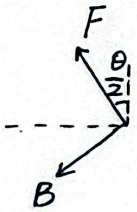


与轴夹角为$d\theta$的长条电流对上述壁筒的磁力dF

$$dF=d(BIL)$$

其中 $B=\frac{\mu_0I'}{2\pi a}=\frac{\mu_0jR\cdot\theta}{2\pi\cdot2R\sin\frac{\theta}{2}}=\frac{\mu_0j\cdot\theta}{4\pi\sin\frac{\theta}{2}}$

$$dI=j\cdot dt$$
$$dL=dl$$
$$dF=\frac{\mu_0j\,d\theta}{4\pi\sin\frac{\theta}{2}}\cdot jdt\cdot dl=\frac{\mu_0j^2\,d\theta\,dt\,dl}{4\pi\sin\frac{\theta}{2}}$$



单位面积的壁筒受到的力

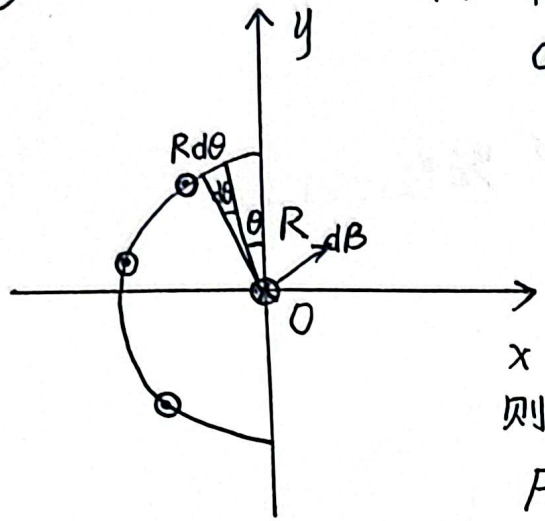
$$dF'=\frac{dF}{dl\cdot dt}=\frac{\mu_0j^2d\theta}{4\pi\sin\frac{\theta}{2}}$$

又由对称性可知整个圆筒对该区域壁筒的磁力沿径向

$$F_0=\int\sin\frac{\theta}{2}\cdot dF'=\int_0^{2\pi}\frac{\mu_0j^2d\theta}{4\pi}=\frac{\mu_0j^2}{2}$$

扫描全能王 创建

18.26



(1) 取 Δl 长度的导线，与轴线夹角为 $d\theta$ 的半圆柱面导体

$$dF = \Delta l \cdot I \cdot dB$$

$$= \Delta l \cdot I \cdot \frac{\mu_0 dI}{2\pi R}$$

$$= \Delta l \cdot I \cdot \frac{\mu_0}{2\pi R} \frac{I \cdot R d\theta}{\pi R}$$

$$= \frac{\mu_0 I \cdot \Delta l}{2\pi R} \cdot \frac{I d\theta}{\pi}$$

则轴线上导线单位长度所受的力

$$F = \frac{\int dF}{\Delta l} = \int_0^{\pi} \frac{\mu_0 I^2}{2\pi^2 R} d\theta = \frac{\mu_0 I^2}{2\pi R}$$

由对称性可知，半圆柱面导体对轴线上的导线的力沿 x 轴正方向

$$F' = \int \sin\theta \frac{dF}{\Delta l} = \int_0^{\pi} \frac{\mu_0 I^2}{2\pi^2 R} \sin\theta \, d\theta$$

$$= \frac{\mu_0 I^2}{\pi^2 R}$$ 方向沿 x 轴正方向

(2) 设两导线之间距离为 d

则其在该导线处产生的磁感应强度

$$B' = \frac{\mu_0 I}{2\pi d}$$

$$B' \cdot \Delta l \cdot I = F' \cdot \Delta l$$ 解 $d = \frac{\pi R}{2}$

扫描全能王 创建

18.27

子弹受的磁力

$$F = BIL = \int_r^{a+r} IB\,dr$$

其中 $B = \frac{2\mu_0 I}{4\pi r}$

$$F = \int_r^{a+r} \frac{2\mu_0 I^2}{4\pi r}\,dr = \frac{\mu_0 I^2}{2\pi} \ln\frac{a+r}{r}$$ 证毕

(1) 子弹的平均加速度 $\bar{a}$

$V^2 = 2\bar{a}L$ 解 $\bar{a} = \frac{V^2}{2L} = \frac{(4.2\times10^3)^2}{2\times5} = 1.76\times10^6\ m/s^2$

$$\frac{\bar{a}}{g} = \frac{1.76\times10^6}{9.8} = 1.8\times10^5$$

即平均加速度为重力加速度的 1.8×10^5 倍

(2) 又 $F = m\bar{a}$

即 $\frac{\mu_0 I^2}{2\pi}\ln\frac{a+r}{r} = m\cdot\frac{V^2}{2L}$

解 $I = \sqrt{\frac{mV^2\cdot\pi}{\mu_0 L \ln\frac{a+r}{r}}} = \sqrt{\frac{317\times10^{-3}\times(4.2\times10^3)^2\cdot\pi}{4\pi\times10^{-7}\times5\times\ln\frac{1.2+6.7}{6.7}}} = 4.12\times10^6\ A$

(3) 子弹获得的能量全部转化为动能

$E = \frac{1}{2}mV^2$

又 $P\eta t = E$

$t = \frac{2L}{V}$ (匀加速运动), $\eta = 40\%$

解 $P = \frac{mV^3}{4L\cdot\eta} = \frac{317\times10^{-3}\times(4.2\times10^3)^3}{4\times5\times40\%} = 2.94\times10^9\ W$

扫描全能王 创建